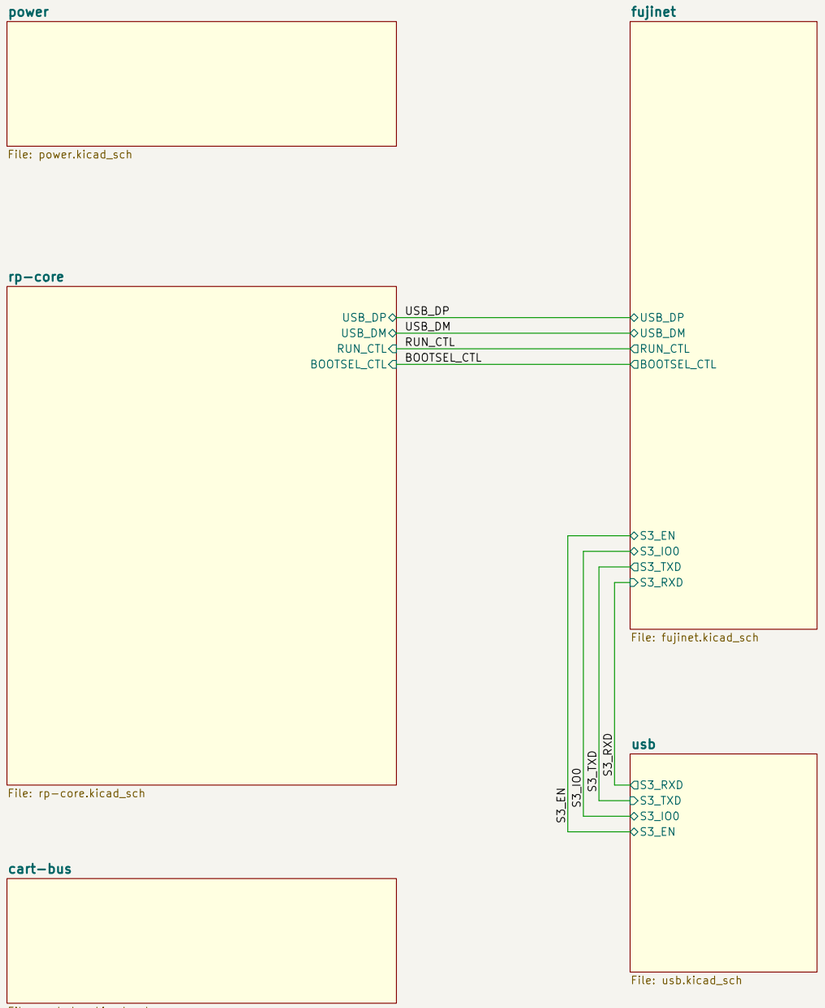
Inty-FujiCard v1.1: Yannick Erb's PINTY-FUJI-CARD, FujiNet for the Intellivision on one cartridge:
an RP2354A on the console's bus + an ESP32-S3.

The cart serves the CP1610's bus in software (fujinet-firmware pico/intellivision, board fujicard); the
S3 is the FujiNet (fujiversal-intv) and the RP's USB host.

cart-bus	the 44-finger edge, the bus straight into GP0-15 / 21 / 22 / 26 / 27, the console reset (GP20, Q1), the ECS audio (GP28, R5), the TX / RX pads
rp-core	the RP2354A's supplies and decoupling, core regulator, crystal, SWD, RUN / BOOTSEL with their buttons and the S3's transistors, its USB out to the S3
fujinet	the ESP32-S3, its EN / BOOT buttons, microSD, SK6805 status LED
usb	USB-C, ESD diodes, the CP2102N bridge and esptool's auto-program pair
power	VIN = the console's +5V or USB VBUS (two SS34), the AP63203 buck: 3V3 for everything

Rails (GND, VSYS, VBUS, VIN, 3V3, 1V1) join by their power symbols.

The board came first: tools/convert.py brought Yannick's EasyEDA layout to KiCad unchanged, and this
schematic is drawn from its netlist (tools/design.py, checked against the board by
tools/board_netlist.py; the board against this schematic by KiCad's DRC parity).
tools/check_nets.py checks this netlist against the firmware (README: firmware).



the console cartridge slot (the 44-finger edge) is on cart-bus

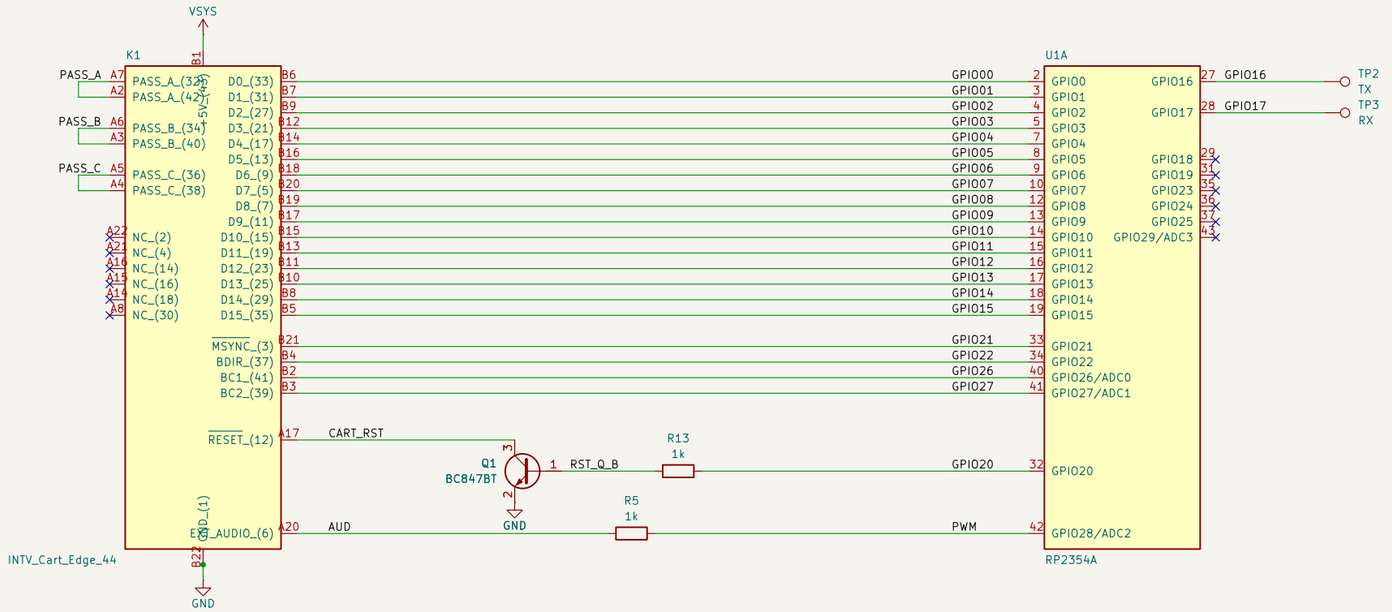
Edge: INTV pin 46-2k on A(k), 45-2k on B(k)
Drawn from the board netlist; RP2354A + ESP32-S3, 2 layers
Design: Yannick Erb (EasyEDA PINTY-FUJI-CARD v1.1)
Generated by tools/gen_sch.py: edit design.py / sch_layout.py
FujiNet

Sheet: /
File: Inty-FujiCard.kicad_sch

Title: Inty-FujiCard v1.1

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 1/6

The Intellivision's cartridge port: 44 contacts, 2 x 22 at 2.54 mm. Yannick's footprint numbers the fingers A1–A22 (component face) and B1–B22 (back), left to right seen from the component face, contacts down: finger A(k) is INTV pin 46 − 2k, B(k) pin 45 − 2k (even pins in front, odd behind); each pin is named with its INTV number.
The CP1610 bus (AD0–15, BDIR, BC1, BC2) and MSYNC reach the RP2354A straight (fujinet-firmware pico/intellivision boards/fujicard.h: MI_AD0_PIN 0, MSYNC 21, BDIR 22, BC1 26, BC2 27, RST 20). GP26 / GP27 (BC1 / BC2) are ADC pins, not 5 V tolerant (GPO–25 are), as on the PiNTY CARD. GP20 high turns Q1 on: the console's /RESET (pin 12) low.
GP28 drives EXT_AUDIO (pin 6, the ECS audio) through 1k (pintycard.h MI_AUDIO_PIN 28). The three shorted pairs pass a console signal through the cart (32 / 42, 34 / 40, 36 / 38). GP16 / GP17 go to the TX / RX pads.



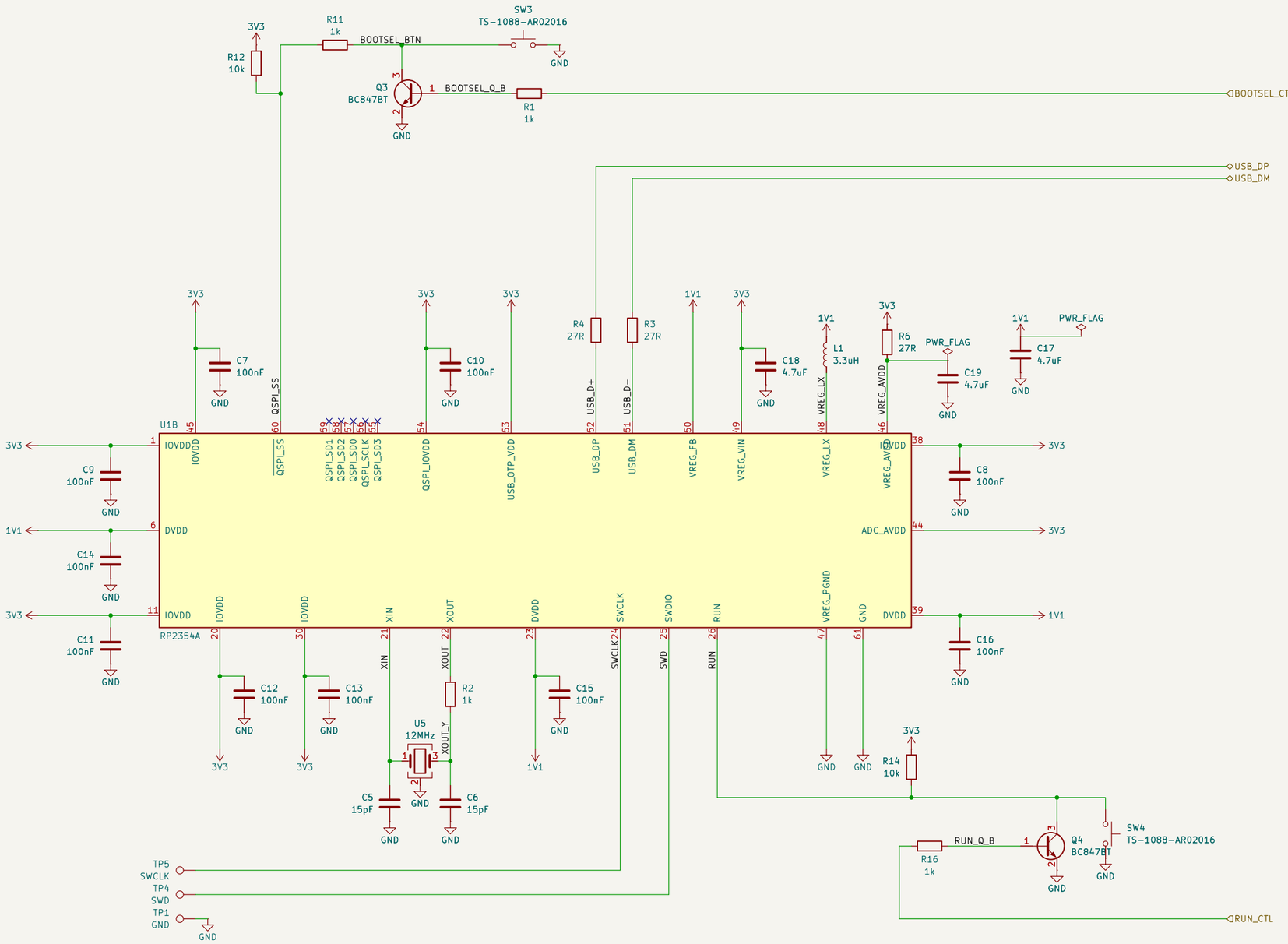
Edge: INTV pin 46−2k on A(k), 45−2k on B(k)
Drawn from the board netlist; RP2354A + ESP32-S3, 2 layers
Design: Yannick Erb (EasyEDA PINTY-FUJI-CARD v1.1)
Generated by tools/gen_sch.py: edit design.py / sch_layout.py
FujiNet

Sheet: /cart-bus/
File: cart-bus.kicad_sch

Title: Inty-FujiCard v1.1 — Cart bus: the edge, RP2354A GPIO

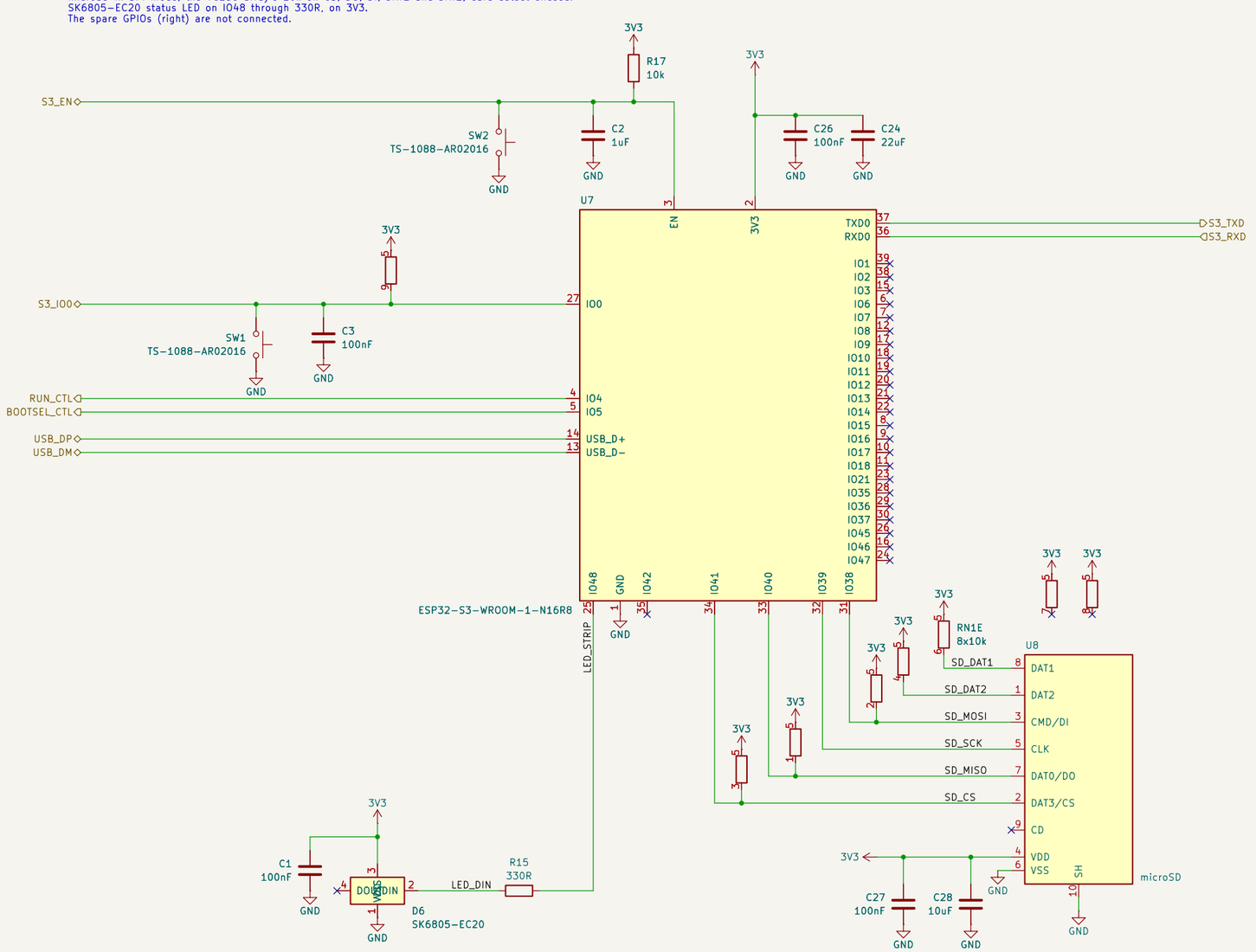
Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 2/6

Supplies (RP2350 datasheet): IOVDD, QSPI_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on 3V3 (the buck, power sheet);
the core regulator switches VREG_LX through 3.3 uH onto 1V1 (VREG_FB senses it), its analog supply VREG_AVDD
through 27R / 4.7 uF. Capacitors as Yannick placed them, each drawn at the pin it serves (C10 at QSPI_IOVDD and
USB_OTP_VDD; ADC_AVDD and USB_OTP_VDD have none of their own). The RP2354A's 2 MB flash is in the package:
its QSPI data and clock pins are not connected; QSPI_SS is the BOOTSEL strap (10k up).
RP-BOOT (SW3) pulls it low through R11; so does Q3 when the S3 raises BOOTSEL_CTL (IO5). RP-RUN (SW4) resets
the RP; so does Q4 when the S3 raises RUN_CTL (IO4): high = asserted (README: firmware). USB: 27R at the RP,
the S3 is the host. SWD and GND on bring-up pads (TP4, TP5, TP1).

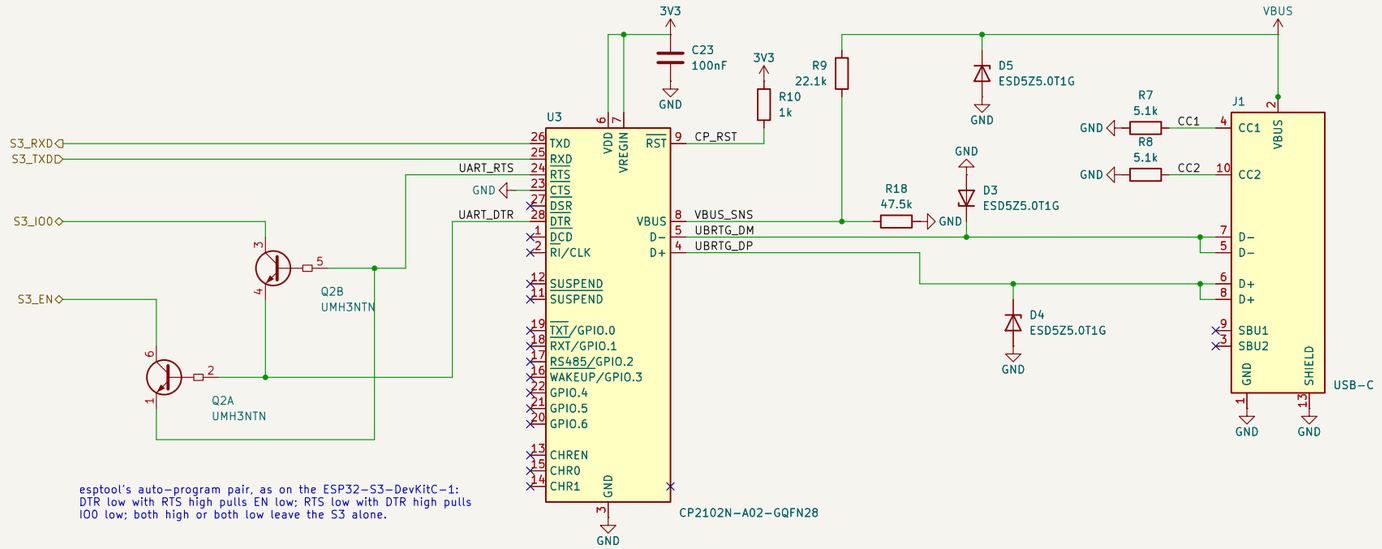


Edge: INTV pin 46–2k on A(k), 45–2k on B(k)	
Drawn from the board netlist; RP2354A + ESP32-S3, 2 layers	
Design: Yannick Erb (EasyEDA PINTY-FUJI-CARD v1.1)	
Generated by tools/gen_sch.py: edit design.py / sch_layout.py	
FujiNet	
Sheet: /rp-core/	
File: rp-core.kicad_sch	
Title: Inty-FujiCard v1.1 – RP2354A core, RUN / BOOTSEL, SWD	
Size: A3	Date: 2026-11-04
KiCad E.D.A. 10.0.8	Id: 3/6

ESP32-S3-WROOM-1-N16R8: the FujiNet (include/pinmap/fujiversal-intv.h), on 3V3 (the buck).
EN: 10k / 1 uF power-on delay; RESET (SW2) and the USB bridge's auto-program pair
(usb sheet) pull it low; BOOT (SW1) and the same pair pull IO0 (100 nF, 10k up).
IO4 / IO5 drive the RP2354A's RUN and BOOTSEL through Q4 / Q3 (rp-core: high = asserted);
IO19 / IO20, the S3's own USB, host the RP's.
microSD in SPI mode, the YC158 array's 10k on CS, DO, DI, DAT1 and DAT2; card detect unused.
SK6805-EC20 status LED on IO48 through 330R, on 3V3.
The spare GPIOs (right) are not connected.



Edge: INTV pin 46-2k on A(k), 45-2k on B(k)	
Drawn from the board netlist; RP2354A + ESP32-S3, 2 layers	
Design: Yannick Erb (EasyEDA PINTY-FUJI-CARD v1.1)	
Generated by tools/gen_sch.py: edit design.py / sch_layout.py	
FujiNet	
Sheet: /fujinet/	
File: fujinet.kicad_sch	
Title: Inty-FujiCard v1.1 – FujiNet: ESP32-S3, microSD, LED	
Size: A3	Date: 2026-11-04
KiCad E.D.A. 10.0.8	Id: 4/6



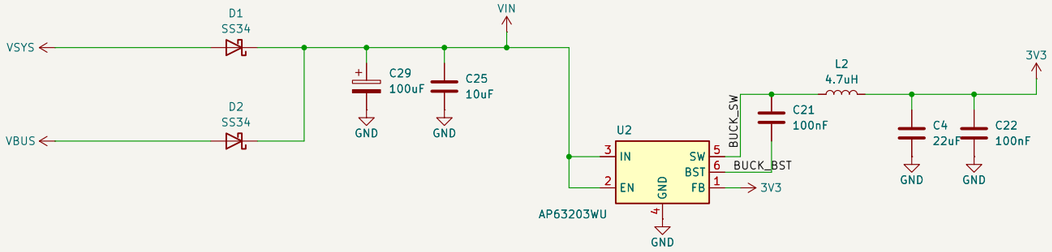
Edge: INTV pin 46−2k on A(k), 45−2k on B(k)
Drawn from the board netlist; RP2354A + ESP32−S3, 2 layers
Design: Yannick Erb (EasyEDA PINTY−FUJI−CARD v1.1)
Generated by tools/gen_sch.py: edit design.py / sch_layout.py
FujiNet

Sheet: /usb/
File: usb.kicad_sch

Title: Inty−FujiCard v1.1 − USB−C + CP2102N

Size: A3	Date: 2026−11−04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 5/6

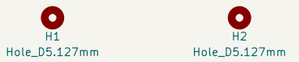
Power tree: VIN is the higher of the console's +5V (VSYS, edge pin 43) and USB VBUS (usb sheet), each through an SS34 Schottky (D1, D2): neither source back-feeds the other, and a cart on USB never powers a console that is off. C29 100 uF and C25 10 uF hold VIN. U2 AP63203, 3.3 V / 2 A synchronous buck (4.7 uH, 22 uF + 100 nF out), makes 3V3, the board's one logic rail: the RP2354A (its core regulator makes 1V1, rp-core), the ESP32-S3, the CP2102N, the microSD and the status LED. VSYS and VBUS are global: the edge (cart-bus) and the USB-C (usb) carry them here.



PWR_FLAG: these rails reach their loads only through passive parts (the edge finger, USB-C, the diodes, the buck's inductor).



H1, H2: the shell's post holes, 5.127 mm NPTH.



Edge: INTV pin 46-2k on A(k), 45-2k on B(k)
Drawn from the board netlist; RP2354A + ESP32-S3, 2 layers
Design: Yannick Erb (EasyEDA PINTY-FUJI-CARD v1.1)
Generated by tools/gen_sch.py: edit design.py / sch_layout.py
FujiNet

Sheet: /power/
File: power.kicad_sch

Title: Inty-FujiCard v1.1 — Power: diode-OR, 3.3 V buck

Size: A3	Date: 2026-11-04	**Rev: 0**
KiCad E.D.A. 10.0.8		Id: 6/6